

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

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Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS sql_db;  
USE sql_db;
```

Output: -

✓	3	09:44:54	CREATE DATABASE IF NOT EXISTS sql_db	1 row(s) affected	0.000 sec
✓	4	09:44:58	USE sql_db	0 row(s) affected	0.016 sec

Query 2: -CREATE PHYSICIAN TABLE

```
CREATE TABLE physician (  
    employeeid INT PRIMARY KEY,  
    name VARCHAR(100),  
    position VARCHAR(100),  
    ssn VARCHAR(20)  
);
```

Output: -

✓	5	09:46:21	CREATE TABLE physician (employeeid INT PRI...	0 row(s) affected	0.063 sec
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Query 3: -INSERT DATA INTO PHYSICIAN

```
INSERT INTO physician VALUES  
(1, 'YISHNAANI', 'STAFF INTERESTS', '111111111'),  
(2, 'PRIYA', 'ATTENDING PHYSICAL', '222222222'),  
(3, 'ARJUN', 'SPORTS ACTIVITIES', '333333333'),  
(4, 'ANANYA', 'FITNESS PROGRAMS', '444444444'),  
(5, 'ROHAN', 'PHYSICAL TRAINING', '555555555'),  
(6, 'SAI', 'YOGA ACTIVITIES', '666666666'),  
(7, 'NEHA', 'STAFF INTERESTS', '777777777'),  
(8, 'VIKRAM', 'ATTENDING PHYSICAL', '888888888'),  
(9, 'KAVYA', 'SPORTS ACTIVITIES', '999999999'),  
(10, 'ADITYA', 'FITNESS PROGRAMS', '1234567890');
```

Output: -

9	14:32:03	INSERT INTO physician VALUES (1,'M.Aswitha P...	9 row(s) affected Records: 9 Duplicates: 0 Wamin...	0.016 sec
---	----------	---	---	-----------

Query 4: -CREATE DEPARTMENT TABLE

```
CREATE TABLE department (  
    departmentid INT PRIMARY KEY,  
    name VARCHAR(100),  
    head INT  
);
```

Output: -

10	14:33:35	CREATE TABLE department (departmentid INT ...	0 row(s) affected	0.047 sec
----	----------	--	-------------------	-----------

Query 5: -INSERT DATA INTO DEPARTMENT

```
INSERT INTO department VALUES  
(1,'General Medicine',4),  
(2,'Surgery',7),  
(3,'Psychiatry',9);
```

Output: -

11	14:35:14	INSERT INTO department VALUES (1,'General Me...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.016 sec
----	----------	---	---	-----------

Query 6: -INNER JOIN

```
SELECT d.name AS Department,  
       p.name AS Head_Physician  
FROM department d  
JOIN physician p  
ON d.head = p.employeeid;
```

Output: -

12	14:36:41	SELECT d.name AS Department, p.name AS ...	3 row(s) returned	0.000 sec / 0.000 sec
----	----------	--	-------------------	-----------------------

Result Grid	Filter Rows:
Department	Head_Physician
General Medicine	Anirudh Reddy
Surgery	Yashwanth Reddy
Psychiatry	Ramya Sree

Query 7: - WHERE CLAUSE

```
SELECT *  
FROM physician  
WHERE position='Surgical Attending Physician';
```

Output: -

13 14:38:23 SELECT * FROM physician WHERE position='Sur... 3 row(s) returned 0.016 sec / 0.000 sec

	employeeid	name	position	ssn
▶	3	Akshara Reddy	Surgical Attending Physician	333333333
	6	Ananya Varshitha	Surgical Attending Physician	666666666
	7	Yashwanth Reddy	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

Query 8: - WHERE + LIKE

```
SELECT *  
FROM physician  
WHERE name LIKE 'Nihitha%';
```

Output: -

15 14:40:11 SELECT * FROM physician WHERE name LIKE '...' 1 row(s) returned 0.000 sec / 0.000 sec

	employeeid	name	position	ssn
▶	8	Nihitha	MD Resident	888888888
*	NULL	NULL	NULL	NULL

Query 9: - COUNT()

```
-- 2520030001-Aswitha  
SELECT COUNT(*) AS Total_Physicians  
FROM physician;
```

Output: -

16 14:42:16 SELECT COUNT(*) AS Total_Physicians FROM p... 1 row(s) returned 0.032 sec / 0.000 sec

	Total_Physicians
▶	9

Query 10: - COUNT DISTINCT

```
SELECT COUNT(DISTINCT position)
FROM physician;
```

Output: -

✓ 17 14:44:11 SELECT COUNT(DISTINCT position) FROM phys... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:
	COUNT(DISTINCT position)	
▶	7	

Query 11: -GROUP BY

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position;
```

Output: -

✓ 18 14:47:37 SELECT position, COUNT(*) AS Total_Emplo... 7 row(s) returned 0.016 sec / 0.000 sec

	position	Total_Employees
▶	Staff Internist	1
	Attending Physician	1
	Surgical Attending Physician	3
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

Query 12: -GROUP BY + HAVING

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

Output: -

✓ 19 14:49:06 SELECT position, COUNT(*) AS Total_Emplo... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid			Filter Rows:	Export:
	position	Total_Employees		
▶	Surgical Attending Physician	3		

Query 13: -ORDER BY

```
SELECT *
FROM physician
ORDER BY name ASC;
```

Output: -

✓ 20 14:50:34 SELECT * FROM physician ORDER BY name AS... 9 row(s) returned 0.032 sec / 0.000 sec

position	ssn
Head Chief of Medicine	555555555
Surgical Attending Physician	333333333
Surgical Attending Physician	666666666
Senior Attending Physician	444444444
Attending Physician	222222222
Staff Internist	111111111
MD Resident	888888888
Attending Psychiatrist	999999999
Surgical Attending Physician	777777777
NULL	NULL

Query 14: -ORDER BY DESC

```
SELECT *
FROM physician
ORDER BY employeeid DESC;
```

Output: -

✓ 20 14:50:34 SELECT * FROM physician ORDER BY name AS... 9 row(s) returned 0.032 sec / 0.000 sec

position	ssn
Head Chief of Medicine	55555555
Surgical Attending Physician	33333333
Surgical Attending Physician	66666666
Senior Attending Physician	44444444
Attending Physician	22222222
Staff Internist	11111111
MD Resident	88888888
Attending Psychiatrist	99999999
Surgical Attending Physician	77777777
NULL	NULL

Query 15: - WHERE + IN

```

SELECT *
FROM physician
WHERE employeeid IN
(
    SELECT head
    FROM department
);

```

Output: -

21 14:54:58 SELECT * FROM physician WHERE employeeid I... 3 row(s) returned 0.000 sec / 0.000 sec

employeeid	name	position	ssn
4	Anirudh Reddy	Senior Attending Physician	44444444
7	Yashwanth Reddy	Surgical Attending Physician	77777777
9	Ramya Sree	Attending Psychiatrist	99999999
NULL	NULL	NULL	NULL

Query 16: -SUBQUERY

```

SELECT *
FROM physician
WHERE employeeid NOT IN

```

```

(
    SELECT head
    FROM department
);

```

Output: -

✓ 22 14:56:50 SELECT * FROM physician WHERE employeeid ... 6 row(s) returned 0.032 sec / 0.000 sec

3	Akshara Reddy	Surgical Attending Physician	333333333
5	Adithya Vardhan	Head Chief of Medicine	555555555
6	Ananya Varshitha	Surgical Attending Physician	666666666
8	Nihitha	MD Resident	888888888
*	NULL	NULL	NULL

Query 17: - GROUP BY + ORDER BY

```

SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
ORDER BY Total_Employees DESC;

```

Output: -

✓ 23 14:58:30 SELECT position, COUNT(*) AS Total_Emplo... 7 row(s) returned 0.000 sec / 0.000 sec

	position	Total_Employees
▶	Surgical Attending Physician	3
	Staff Internist	1
	Attending Physician	1
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

Query 18: - JOIN + WHERE

```
SELECT p.name
FROM physician p
JOIN department d
ON p.employeeid=d.head
WHERE d.name='Surgery';
```

Output: -

✓ 24 15:01:13 SELECT p.name FROM physician p JOIN depart... 1 row(s) returned 0.000 sec / 0.000 sec

	name
▶	Yashwanth Reddy

Query 19: - JOIN + ORDER BY

```
SELECT d.name,
       p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
```

Output: -

✓ 25 15:02:46 SELECT d.name, p.name FROM department ... 3 row(s) returned 0.000 sec / 0.000 sec

	name	name
►	General Medicine	Anirudh Reddy
	Psychiatry	Ramya Sree
	Surgery	Yashwanth Reddy

Query 20: - WHERE + NOT LIKE

```
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
```

Output: -

✓ 26 15:04:17 SELECT * FROM physician WHERE position NO... 6 row(s) returned 0.000 sec / 0.000 sec

position	ssn
Staff Internist	111111111
Attending Physician	222222222
Senior Attending Physician	444444444
Head Chief of Medicine	555555555
MD Resident	888888888
Attending Psychiatrist	999999999
NULL	NULL

Query 21: - COUNT + HAVING

```
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

Output: -

✓ 27 15:05:45 SELECT position, COUNT(*) AS Total FROM ... 1 row(s) returned 0.000 sec / 0.000 sec

	position	Total
►	Surgical Attending Physician	3

Query 22: - EXISTS

```

HAVING COUNT(*) >= 4;
• SELECT *
  FROM department d
 WHERE EXISTS
  (
    SELECT 1
    FROM physician p
    WHERE p.employeeid=d.head
  );

```

Output: -

✓ 28 15:07:03 SELECT * FROM department d WHERE EXISTS ... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid




Filter Rows:

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
•	NULL	NULL	NULL

Query 23: - MAX()

```

SELECT MAX(employeeid)
FROM physician;

```

Output: -

✓ 29 15:08:41 SELECT MAX(employeeid) FROM physician LIMIT... 1 row(s) returned 0.000 sec / 0.000 sec

MAX(employeeid)
9

Query 24: - MIN()

```

SELECT MIN(employeeid)
FROM physician;

```

Output: -

✓ 30 15:10:24 SELECT MIN(employeeid) FROM physician LIMIT... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	
	MIN(employeeid)
▶	1